

Project Context Cheat Sheet

CMS · Molina · Providers · Patients — What this project is actually about

1. CMS runs two different programs

CMS (Centers for Medicare & Medicaid Services) is the federal agency that pays for healthcare for low-income and elderly Americans. It runs two types of programs, and they work very differently:

Program	How it works
Medicare Fee-for-Service (FFS) ← YOUR DATASET	CMS pays providers (doctors, hospitals) directly per claim submitted.
Medicaid / Medicare Advantage ← MOLINA'S WORLD	CMS pays Molina a fixed monthly amount per member. Molina then pays providers from the

2. How money flows in YOUR dataset (CMS Medicare FFS)

Step by step, what happens when a Medicare patient sees a provider:

- Patient sees a doctor or goes to a hospital
- Provider sends a bill to CMS — e.g., \$500
- CMS pays the provider less than billed — e.g., \$180
- Provider is required by law to accept \$180 as full payment. They cannot bill anyone else for the \$320 gap.
- The \$320 gap is absorbed by the provider as a loss. Molina is not involved. The patient doesn't pay it.

This gap — what providers bill vs. what CMS pays — is what your pipeline measures. Your dashboard surfaces which provider types and service categories have the widest gap.

3. How money flows in Molina's world (Medicaid / Medicare Advantage)

- CMS/state pays Molina a fixed monthly amount per enrolled member (called a capitation payment)
- Member goes to a doctor, hospital, or pharmacy
- Provider sends the bill to Molina
- Molina pays the provider based on a negotiated contract rate
- If Molina's total claims costs exceed what CMS paid them → Molina absorbs the loss out of its own margin

Molina's entire financial health depends on keeping claims costs below capitation payments. This is measured by the Medical Cost Ratio (MCR) — the % of revenue paid out in claims. The lower the MCR, the healthier Molina's margins.

4. So why would Molina care about CMS FFS payment gaps?

Three legitimate reasons:

Reason 1 — Provider contracts are benchmarked to Medicare rates.

Molina doesn't pay hospitals whatever they ask. Contracts are negotiated as a percentage of Medicare rates — e.g., 'we pay this hospital 120% of what Medicare pays.' If Medicare pays \$180 for a procedure, Molina pays \$216. Knowing where Medicare underpays tells a Molina analyst exactly where their own contracted costs are concentrated.

Reason 2 — Denial patterns are the same everywhere.

Claims get denied for the same reasons regardless of payer — wrong codes, missing prior authorization, medical necessity gaps. The denial patterns in CMS FFS data directly reflect the same patterns Molina's analysts deal with on their internal Medicaid and Medicare Advantage claims.

Reason 3 — Risk signals transfer.

Providers who aggressively overbill Medicare also aggressively overbill Molina. Your data surfaces which provider types and service categories carry the highest billing-to-payment gap — those are the same risk signals a Molina claims analyst would flag in internal data.

5. Your project in plain English

You used the public CMS Medicare FFS dataset because it is the best publicly available claims dataset. You built a pipeline that surfaces WHERE the payment gap is widest — by provider type, service type, and location. The skills you used (SQL, dbt, Snowflake, star schema, Streamlit dashboard) are exactly what a Molina claims analyst uses on Molina's internal claims data every day. The dataset is public. The skills and insights transfer directly.

The one-line answer if the professor asks:

"I used public CMS Medicare data to demonstrate claims analytics skills — the payment gap patterns I surfaced are directly relevant to how Molina benchmarks its provider contracts and manages claim denial risk."